Second-Order Transimpedance Amplifiers in Mixer-First Receivers: Design for Optimum Blocker Tolerance

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***Abstract*— A design-oriented analysis of a transimpedance amplifier (TIA) reveals the optimum compression-free dynamic range for downconverted blockers lying in its 2nd-order transition band. The unique circuit topology is chosen to eliminate low-frequency zeros of the transfer function to TIA output with only one OpAmp. Two approaches lead to TIA designs with or without capacitive positive feedback. Both configurations are analyzed and discussed. Simple and accurate design equations allow the component value to be selected straightforwardly. ω_0 and Q of the two poles are independently controlled. Analytical expressions for out-of-band (OOB) linearity are derived, explaining different distortion mechanisms from both gate and drain voltages of a FET. The FET gate voltage is shown to affect the distortion even if its swing is much lower than the drain voltage swing. The design fully benefits from 2nd-order filtering, yielding optimum blocker tolerance. Power dissipation is limited by the requirements for noise figure and independent pole control. A mixer-first receiver that operates from 1 to 7 GHz is realized on 65 nm RF-SOI CMOS. It displays +31.1 dBm OOB input-referred third-order intercept point (OOB-IIP3), and +11.8 dBm OOB blocker 1-dB compression point (B1dB). The TIA consumes 22 mW. The prototype verifies the TIA design.**

***Index Terms*— Mixer-first receiver, baseband, second-order transimpedance amplifier (TIA), filters, positive feedback, blocker tolerance, linearity, single amplifier biquad (SAB).**

I. Introduction

MULTIBAND receivers (RXs) suffer interference from out-of-band (OOB) blockers in a number of scenarios, including leakage from their own transmitter (TX) when operating in frequency-division duplex (FDD). With a large number of bands, the traditional use of bulky, fixed off-chip SAW filters becomes impractical. Nor can SAW filters reject strong in-band (IB) blockers. Receiver circuits must be designed to tolerate the largest possible unfiltered blocker before it forces them into gain compression, raises the noise figure and produces unacceptable intermodulation.

A mixer-first receiver offers higher linearity. As shown in Fig. 1a, this receiver shifts the frequencies received at the antenna to baseband (BB) without amplification at RF. The LO frequency is tuned to position the wanted channel at zero IF. The mixer takes its input directly from the antenna through a passive matching network. The bilateral passive mixer downconverts the RF signal to baseband, while at the same time upconverting BB signals to RF [1]. Therefore, the BB impedance, and its filtering effect, are translated to RF.

The active baseband transimpedance circuit combines amplification with filtering. But this filtering is not ideal: it is gradual. Close-in blockers are filtered mildly, far-away blockers more so. The rate of filter rolloff, that is the slope of its transition band with frequency, should be steep. Close-in blockers can drive the FETs in this circuit into their nonlinear region, and create wideband intermodulation distortion that may overlap the wanted channel.

As substantial a blocker suppression as is possible should be obtained with the fewest transistors: a partly-filtered blocker can force any transistor into unacceptable nonlinear operation. [2], [3], [4] have shown that up to a 2nd-order lowpass filter can be constructed with *one* ideal voltage OpAmp and an *RC* network, using a mix of positive and negative feedback to create complex poles for a maximally flat response. [5], [6] unify designs for these single amplifier biquads (SABs). The central problem in a wireless receiver is that the dynamic range of the active filter is tested by large blockers that lie in the filter's transition band or stopband. Intermodulation is then strongly dependent on the filter's frequency response, a subject that has not been tackled in the well-known analyses of distortion in transistor circuits. When compared to a transimpedance amplifier (TIA), classic 2nd-order SABs constructed with ideal voltage OpAmps require additional resistors at their input stage. This worsens their noise figure (NF).

Fig. 1a shows a traditional TIA[1] with 1st-order lowpass filter following the passive mixer. Enhanced baseband selectivity has been studied in mixer-first receivers [7], [8], [9], [10], [11], [12], [13], [14], [15], whose circuit topologies can be put into three categories based on the equivalent circuit at zero IF and number of OpAmps needed:

1) Traditional TIA with one OpAmp: [7], [8], [9] (Fig. 1b-1d).
2) Traditional TIA with more than one OpAmp: [10], [11], [12], [13] (Fig. 1e-1f).
3) Other amplifier topologies with more than one OpAmp: [14], [15].

Some of these papers treat as their primary objective the realization of 2nd-order filtering with a single OTA/OpAmp, relegating blocker tolerance as a secondary outcome. To us, the existing literature is unsatisfactory in the following respects:

1) A usable design methodology is absent in [7], leading to complicated design, control, and non-optimum power

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[1] An ideal TIA presents low input resistance. A TIA with comparable input resistance to source resistance (e.g. for impedance match) is a non-ideal TIA.

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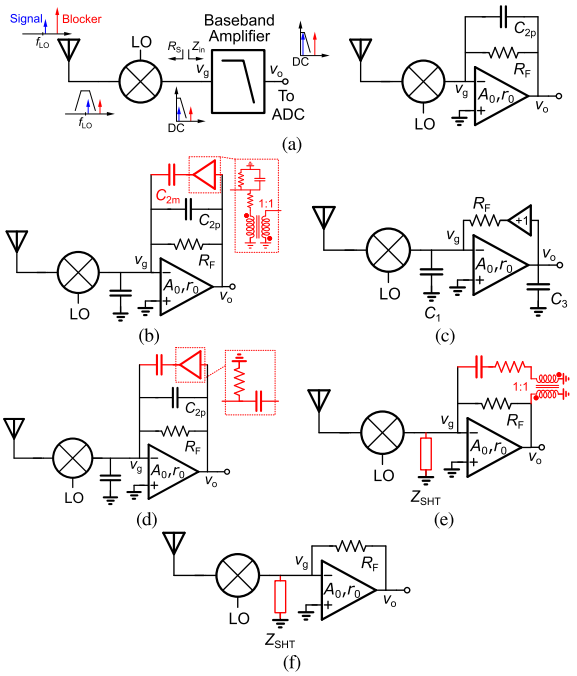


Fig. 1. (a) Mixer-first receiver and traditional 1st-order transimpedance amplifier (TIA). Higher-order transimpedance amplifier in (b) [7], (c) [8], (d) [9], (e) [10], (f) [11], [12], [13], where inverting ideal transformer represents positive feedback connection in differential circuits.

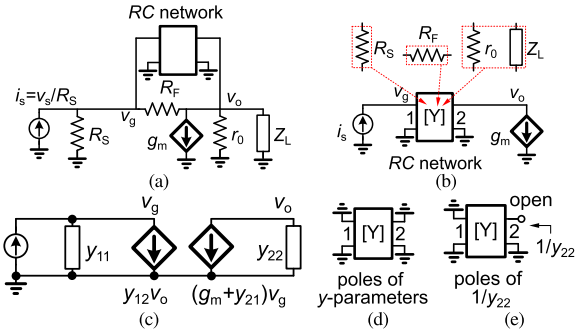


Fig. 2. (a) TIA small signal equivalent circuit with any filtering *RC* network; (b) Simplified two-port *RC* network for analysis purposes and (c) its *y*-parameter equivalent circuit; (d) Equivalent circuit giving poles of *y*-parameters ($\mathbf{I} = \mathbf{YV}$); (e) Equivalent circuit giving poles of y_{22}^{-1}, or zeros of y_{22}.

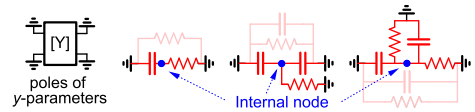


Fig. 3. Feedback network equivalent circuits of (from left to right) [10], [9], [7] giving poles of *y*-parameters and zeros of transfer function $V_o/V_s(s)$.

consumption. In [9], impractical assumptions over-simplify the design, giving inaccurate expressions, and complicating the design process.

2) In [7], [9], [10], [14], the presence of one or two low-frequency zeros in the transfer functions limit 2nd-order filtering to small frequency offsets. Beyond the zero frequency, the rolloff relaxes to first order.
3) Linearity metrics do not fully benefit from the enhanced selectivity in [9], [11], [12], [13], [14]. In [9] for example, IIP3 improves at the rate of only ∼20 dB/decade over the frequency range where the magnitude of the baseband filter is rolling off at 40 dB/decade.
4) [10], [11], [12], [13], [14], [15] use more than one amplifier, wasting power or complicating design.

We will describe remedies for all these shortcomings. This paper concentrates on single-OTA 2nd-order TIAs for their low sensitivity compared to SABs. Our contributions are as follows:

1) We show the *one* TIA circuit topology using an OTA that eliminates low-frequency zeros at the TIA's output, resulting in a wide frequency range with enhanced selectivity.
2) We present comprehensive, accurate, yet simple and intuitive circuit analysis. We arrive at simple, low-entropy design equations, which specify a clear design sequence to satisfy in-band (IB) and out-of-band (OOB) specifications. Among other benefits, ω_0 and Q of the 2nd-order TIA are controlled independently.
3) We develop analytical expressions for distortion, showing that the input port of the TIA can limit the overall linearity even when the voltage swing there is lower than the output voltage swing. We offer methods of design to adjust this. This leads to an optimum design for blocker tolerance, which fully benefits from enhanced selectivity.
4) We show how to lower power consumption—to an extent—without sacrificing simplicity of design and tunability.

In view of the shortcomings of the existing literature, we place blocker tolerance at the forefront of the development that follows.

II. Zeros in Single-OpAmp TIAs

Some reported 2nd-order TIAs [7], [9], [10] suffer from low-frequency zeros. At frequencies beyond that zero, the circuit suppresses blockers in its transition band with only a 1st-order rolloff in its magnitude response. In this section, we discuss the origins of low-frequency zeros, and how to push them out in frequency so that far-away blockers, which are often the largest of all, are attenuated as effectively as is possible. This is where SAW filters excel, but where active filters as their replacements struggle.

The simplest CMOS TIA contains one OTA driven by a passive mixer. A feedback resistor R_F shunts the amplifier input to its output. A capacitor in feedback will create a lowpass characteristic, without disturbing in-band properties such as gain and noise.

Fig. 2a shows a general small-signal equivalent circuit. For purposes of analysis, source resistor R_S, TIA loading Z_L, OTA output resistance r_0, and feedback resistor R_F, can all be absorbed into the *RC* network, as shown in Fig. 2b. Thus a reciprocal *RC* network captures everything except the voltage-controlled current source g_m that models an ideal OTA. The standard *y*-parameter equivalent circuit of the complete TIA is shown in Fig. 2c. y_{12} signifies the strength of the shunt-shunt feedback. When compared to [2], [3], [4], [5], [6], we do not assume an ideal voltage OpAmp, but instead adhere to a representative OTA in CMOS. The results are revealing of limitations and ultimately prove useful for design.

Two transfer functions are considered. The first is from the source v_s to the amplifier/filter *output port* voltage, v_o. The second is to the amplifier *input port* voltage v_g. If the response to a blocker present at the mixer output i_s, or equivalently v_s, causes some voltage limit to be exceeded at *either* port, it can drive the entire OTA into gain compression.

The transfer functions are now derived systematically. The unilateral transimpedance $R_{T,o}(s)$ in open loop, input port driving point impedance $Z_{g,o}(s)$ in open loop, and the loop

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gain $T(s)$ are:

$$R_{T,o}(s) = \left.\frac{V_o}{I_s}(s)\right|_{y_{12}=0} = -\frac{y_{21}+g_m}{y_{11}y_{22}}(s) \tag{1a}$$

$$Z_{g,o}(s) = \left.\frac{V_g}{I_s}(s)\right|_{y_{12}=0} = y_{11}^{-1}(s) \tag{1b}$$

$$T(s) = R_{T,o}y_{12} = -\frac{(y_{21}+g_m)}{y_{22}}\frac{y_{12}}{y_{11}}(s). \tag{1c}$$

Using (1a) to (1c), linear feedback theory allows us to write expressions for the *closed-loop* voltage transfer functions:

$$\frac{V_o}{V_s}(s) = \frac{1}{R_S}\times\frac{V_o}{I_s}(s) = \frac{1}{R_S}\times\frac{R_{T,o}}{1+T}(s) \tag{2a}$$

$$\frac{V_g}{V_s}(s) = \frac{1}{R_S}\times\frac{V_g}{I_s}(s) = \frac{1}{R_S}\times\frac{Z_{g,o}}{1+T}(s). \tag{2b}$$

Zeros and poles of the transfer function are, respectively, roots of the polynomials in the numerator and denominator. For all y-parameters $y_{ij}(s) \equiv \frac{y_{ij,N}(s)}{\Delta_y(s)}$, $i, j = 1, 2$, where $y_{ij,N}(s)$ and $\Delta_y(s)$ are polynomials. Dimensionless $\Delta_y(s)$ is the same for all y-parameters since they must share the same natural modes. These are the natural modes of the equivalent circuit in Fig. 2d, found in the currents induced by initial conditions when both ports are shorted. (2a) and (2b) are then

$$\frac{V_o}{V_s}(s) = \frac{-1}{R_S}\times\frac{(y_{21,N}+g_m\Delta_y)\Delta_y}{y_{11,N}y_{22,N}-y_{12,N}(y_{21,N}+g_m\Delta_y)}(s) \tag{3a}$$

$$\frac{V_g}{V_s}(s) = \frac{1}{R_S}\times\frac{y_{22,N}\Delta_y}{y_{11,N}y_{22,N}-y_{12,N}(y_{21,N}+g_m\Delta_y)}(s). \tag{3b}$$

The numerator and denominator of these expressions for transfer functions are now each linear combinations of products of polynomials. The closed-loop zeros and poles are the roots of these linear combinations.

The transfer function to TIA *output* (3a) may contain zeros that can arise in two different ways:

1) When $g_m = -y_{21}(s)$ in (1a) and (3a). There are two ways to remove the low-frequency zeros. (i) By forcing $y_{21}(s)$ to become independent of s, that is, a constant. For example, no capacitor is present in feedback around the OTA [8], [16]. (ii) By using controlled capacitive *positive* feedback to push the zero to some high enough frequency that it no longer matters. We discuss this later.
2) When $\Delta_y(s) = 0$ in (3a). To eliminate zeros, $\Delta_y(s)$ should be a constant independent of s, so its roots which are zeros in the transfer function lie at $s \to \infty$. Eliminating all finite poles in the configuration of Fig. 2d, i.e. forcing the pole frequencies to lie at ∞, eliminates zeros from $\Delta_y(s)$. This requires that the circuit graph should consist of only two independent signal nodes with respect to ground: the OTA input, and the OTA output. To satisfy this, each two-terminal capacitor or resistor in the *RC* network (Fig. 2b) must connect between TIA input and output nodes, or from either node to ground. But in an ideal voltage OpAmp, this topology of connections will force the transfer function to the TIA output into 1st-order. Therefore, in voltage OpAmps it becomes necessary to introduce a second independent node in the feedback network [2], [3], [4], [5], [6], which means that the two poles *must* be accompanied by one or more zeros. Designers using OTAs who take their cue from ideal OpAmp circuits and construct feedback networks with internal nodes will unwittingly compromise filtering effectiveness because of these zeros. As shown in Fig. 3, the feedback networks in [7], [9], [10] contain internal nodes, and consequently, low-frequency zeros.

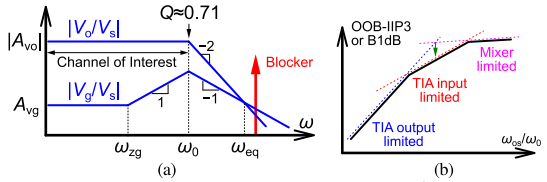


Fig. 4. (a) Typical transfer functions of a 2nd-order TIA without low-frequency zeros in $V_o/V_s(s)$, but with a low-frequency zero (ω_{zg}) in $V_g/V_s(s)$; (b) Receiver OOB-IIP3 (or blocker 1-dB compression point, B1dB) versus blocker offset frequency (ω_{os}), where TIA input can dominate distortion even without zeros in $V_o/V_s(s)$.

The transfer function to TIA *input* (3b) also contains zeros arising from $y_{22,N}$. They are the poles of the driving point impedance y_{22}^{-1} (Fig. 2e). These cannot be eliminated. A zero at a low frequency causes peaking to appear in the transfer function V_g/V_s to the TIA's input port, close to the corner frequency of the passband [8] (Fig. 4a).

For 2nd-order TIAs, Fig. 4a illustrates a case where the magnitude of two transfer functions is equal at ω_{eq}. The amplitude of a blocker at $\omega > \omega_{eq}$ will be larger at the amplifier input than at its output. If the transfer function to output $V_o/V_s(s)$ consists *only* of a complex conjugate pole pair with $Q = 1/\sqrt{2}$ but no zeros for a flat lowpass response, then the transfer function to input, $V_g/V_s(s)$, must consist of the same two poles, *and* a low frequency zero.[2] This means that for impedance match near DC, the transfer function $|V_g/V_s(0)| \approx 1/2$; but because of the low frequency zero ω_{zg}, it rises at 20 dB/dec as frequency approaches the lowpass cutoff; then at frequencies beyond, it falls at 20 dB/dec. Whereas the magnitude of the transfer function to the output $|V_o/V_s|$ is flat until the filter cutoff, then falls at 40 dB/dec. In simple words, at its output port the amplifier appears as a second-order filter for OOB blockers; but at its input port it appears to these blockers as *only* a first-order filter.

We say the dynamic range of this RX front-end is optimum when an input blocker located at a certain offset frequency forces the mixer, TIA input port and output port *all three* to enter into compression, or contribute just unacceptable 3rd-order distortion—whichever happens first. That is, in the optimal circuit the three asymptotic lines in Fig. 4b should all intersect at one frequency.

III. Single-OpAmp TIAs Without Zeros of $\Delta_y(s)$

Fig. 5a shows the only topology of a single-OpAmp TIA where $\Delta_y(s)$ will not contribute zeros. C_{2p} and R_F provide negative feedback as is customary in traditional low-pass TIAs. C_{2m} appears in the feedback with a transformer inversion indicating positive feedback. In a fully differential OpAmp, C_{2m} would be cross-connected to the non-inverting output. The single-ended small signal equivalent circuit is shown in Fig. 5b. The lumped capacitors C_1 and C_3 include the loading of the preceding and succeeding circuit blocks, the capacitance of the amplifier, and switched capacitor banks. R_S and R_L model the source and load resistance. The circuit is second-order because it comprises only two independent nodes.

[2]All legitimate transfer functions where the stimulus is an independent source share the same poles, or natural modes. Transfer functions from an independent source (input) to two different outputs are distinguishable only by their zeros.

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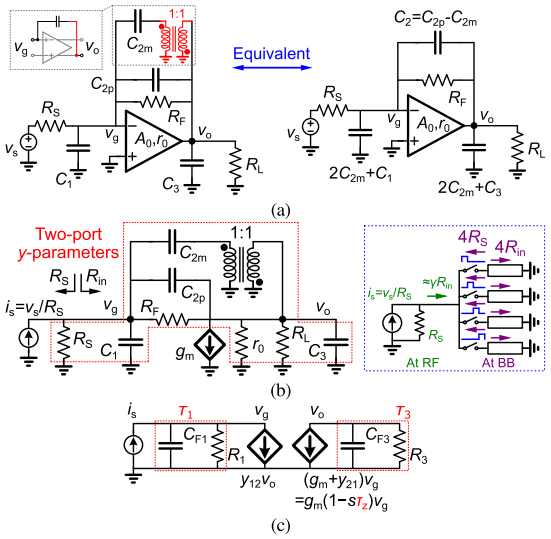


Fig. 5. (a) TIA without zeros from $\Delta_y(s)$ (left) and its equivalence (right), where inverting ideal transformer represents positive feedback in differential circuits; (b) Single-end TIA small signal equivalent circuit for y-parameters analysis; (c) y-parameters equivalent circuit of the TIA.

Suppressing g_m, the parameters for the equivalent circuit are

$$y_{11}(s) = \frac{1}{R_F \parallel R_S} + s(C_{2p} + C_{2m} + C_1) \equiv \frac{1}{R_1} + sC_{F1} \quad (4a)$$

$$y_{12}(s) = -\left(\frac{1}{R_F} + s(C_{2p} - C_{2m})\right) \equiv -\left(\frac{1}{R_F} + sC_2\right) \quad (4b)$$

$$y_{21}(s) = y_{12}(s) \text{ (the network is reciprocal)} \quad (4c)$$

$$y_{22}(s) = \frac{1}{R_F \parallel r_0 \parallel R_L} + s(C_{2p} + C_{2m} + C_3) \equiv \frac{1}{R_3} + sC_{F3}, \quad (4d)$$

where $R_1 = R_S \parallel R_F$, $C_{F1} = C_1 + C_F$ loads the input of the equivalent circuit, $R_3 = R_L \parallel R_F \parallel r_0$, $C_{F3} = C_3 + C_F$ loads the output of the equivalent circuit, r_0 is the amplifier output resistance, and

$$C_2 = C_{2p} - C_{2m} \text{ is the net capacitance in feedback,} \quad (5a)$$

whereas $C_F = C_{2p} + C_{2m}$ loads both input and output. (5b)

C_2 and C_F in (5) have distinct physical meanings. Again, C_2 may be negative. C_{2p} and C_{2m} are selected individually so that their difference defines the feedback capacitance, and their sum the loading at the amplifier ports. Fig. 5c shows the simplified equivalent circuit. y_{11} and y_{22} are the shunt RCs loading the input and output (Fig. 5c), which define the time constants

$$\tau_1 \triangleq R_1 C_{F1}, \qquad \tau_3 \triangleq R_3 C_{F3}. \quad (6)$$

From (1c), the loop gain of the feedback circuit is:

$$T(s) = -\left(R_1 || \frac{1}{sC_{F1}}\right) \times G'_m(s) \times \left(R_3 || \frac{1}{sC_{F3}}\right) \times y_{12}(s)$$
$$\equiv -Z_1(s) G'_m(s) Z_3(s) y_{12}(s), \quad (7)$$

where $Z_1(s) = R_1/(1 + s\tau_1)$, $Z_3(s) = R_3/(1 + s\tau_3)$, and the effective transconductance $G'_m(s) = g_m + y_{21}(s) \approx g_m - sC_2 \triangleq g_m(1 - s\tau_z)$, assuming $R_F \gg 1/g_m$.

The transfer functions are, from (2)(4)(6)(7):

$$\frac{V_o}{V_s}(s) = \frac{-R_1 R_3}{R_S} \times \frac{G'_m(s)}{\Delta_{TF}(s)} = \frac{A_{vo}(1 - \frac{s}{\omega_{zo}})}{1 + \frac{s}{\omega_0 Q} + \frac{s^2}{\omega_0^2}} \quad (8a)$$

$$\frac{V_g}{V_s}(s) = \frac{R_1}{R_S} \times \frac{1 + s\tau_3}{\Delta_{TF}(s)} = \frac{A_{vg}(1 + \frac{s}{\omega_{zg}})}{1 + \frac{s}{\omega_0 Q} + \frac{s^2}{\omega_0^2}}, \quad (8b)$$

where polynomial $\Delta_{TF}(s) \triangleq \big(1 + T(s)\big) \times (1 + s\tau_1)(1 + s\tau_3)$. Each transfer function in (8) contains two poles and one zero. The poles are the roots of the denominator $\Delta_{TF}(s)$. The zeros of the two transfer functions are the roots of the numerator $G'_m(s)$ and $(1 + s\tau_3)$:

$$\omega_{zo} = \frac{g_m - 1/R_F}{C_2} \approx \frac{1}{\tau_z} = \frac{g_m}{C_2} \quad (9a)$$

$$\omega_{zg} = \frac{1}{\tau_3}. \quad (9b)$$

Over the low-frequency passband, all capacitors appear as an open circuit. The feedback amplifier's DC transfer functions A_{vo}, A_{vg} and input resistance R_{in} are as follows:

$$R_{in} = \frac{R_F + r_0 || R_L}{g_m(r_0 || R_L)} \approx \frac{R_F}{g_m r_0} = \frac{R_F}{A_0} \quad (10a)$$

$$A_{vo} = \frac{V_o}{V_s}(0) = -\frac{R_F - R_{in}}{R_S + R_{in}} \approx -\frac{R_F}{R_S + R_{in}} \quad (10b)$$

$$A_{vg} = \frac{V_g}{V_s}(0) = \frac{R_{in}}{R_S + R_{in}}, \quad (10c)$$

where A_0 is the standalone OTA's DC voltage gain when unloaded at its output. (10) assumes $R_F \gg r_0$, $R_L \gg r_0$, and $A_0 \gg 1$. These assumptions will hold true in a good voltage amplifier.

The frequency ω_{eq} where the magnitudes of the two transfer functions are equal in Fig. 4a is calculated from (8)(9)(10) when $|\omega_{zo}| \gg \omega_{eq} \gg \omega_{zg}$:

$$\omega_{eq} \approx \frac{|A_{vo}|}{A_{vg}} \omega_{zg} \approx A_0 \omega_{zg} = \frac{A_0}{\tau_3}. \quad (11)$$

The two poles in (8) are the roots of the polynomial $\Delta_{TF}(s)$:

$$\Delta_{TF}(s) = \left(1 + \frac{R_1 R_3 g_m}{R_F}\right) + \left(R_1 C_{F1} + R_3 C_{F3} + C_2 R_1 R_3 \left(g_m - \frac{2}{R_F}\right)\right)s + \left(R_1 R_3 \big(C_{F1} C_{F3} - C_2^2\big)\right)s^2$$
$$\approx k_{in} + (\tau_1 + \tau_3 + \tau_2)s + \big(\tau_1 \tau_3 - \tau_2 \tau_z\big)s^2, \quad (12)$$

where $\tau_2 \triangleq C_2 R_1 R_3 g_m, \qquad \tau_z = C_2/g_m,$ (13)

and $k_{in} \triangleq 1 + T(0) = 1 + (R_1 R_3 g_m / R_F)$
$\approx 1 + (R_S/R_{in})$ when $R_F \gg R_S$, i.e. $|A_{vo}| \gg 1$. (14)

The expression for the two poles cast into standard form is

$$\Delta_{TF}(s) = k_{in}\left(1 + \frac{s}{\omega_0 Q} + \frac{s^2}{\omega_0^2}\right). \quad (15)$$

ω_0 and Q are found by comparing (12) and (15):

$$\omega_0^2 = \frac{k_{in}}{\tau_1 \tau_3 - \tau_2 \tau_z} \quad (16a)$$

$$Q^{-1} = \frac{\tau_1 + \tau_2 + \tau_3}{\sqrt{k_{in}(\tau_1 \tau_3 - \tau_2 \tau_z)}}. \quad (16b)$$

Once again, the assumptions used to derive (10) and (16) are:

$$g_m R_S \gg 1; \quad A_0, A_{vo} \gg 1; \quad R_L, R_F \gg r_0. \quad (17)$$

In a mixer-first receiver, the TIA input resistance upconverted to RF should equal 50 Ω to match the nominal antenna impedance. According to [1] [17, (39)] a factor of $\gamma = \text{sinc}^2(1/4) \approx 0.81$ will scale the total[3] R_{in} as it appears

[3]This means that the baseband TIA is analyzed with scaling as if all paths are connected in parallel (Fig. 5b). For example, if each TIA path loads a four-path mixer with 200Ω, then total $R_{in} = 50\Omega$ by definition above; this differs from, but is equivalent to, $\gamma = 0.203 = (0.81/4)$ in [1] which uses $R_{in} = 200\Omega$. Both lead to the identical conclusion.

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at a four-path mixer input transformed from the mixer output (Fig. 5b). In practice, impedance match is considered acceptable if $|s_{11}| < -10\,\text{dB}$, which means that the receiver input impedance can range over $0.5R_S$ to $2R_S$. Given this wide range, we can safely round off $\gamma \approx 1$ and enforce $R_{in} = R_S$ for acceptable impedance match. The mixer switch resistance R_{sw} does not matter because by design it is usually $\ll 50\,\Omega$ for low noise and distortion. Perfect impedance match requires slightly different R_{in} and R_S because $\gamma \neq 1$. When the inequalities $g_m R_S \gg 1$ and $A_0 \gg 1$ hold true, then at impedance match, $R_F \gg r_0$ and $|A_{vo}| \gg 1$ because $R_F \approx A_0 R_S = g_m R_S r_0 \gg r_0, R_S$. Now a smaller set of constraints applies:

$$g_m R_S \gg 1;\;\; A_0 \gg 1;\;\; R_L \gg r_0. \tag{18}$$

If the third constraint is not met, then after replacing r_0 with $r_0 \parallel R_L$ wherever the approximation is used, the small signal expressions will remain valid. The constraints (18) fail only for a small g_m.

(16) shows that in an OTA τ_3 at the output node plays an equal role to τ_1 in defining filter poles. [2], [3], [4] assume an ideal voltage OpAmp ($\tau_3 \to 0$). Their design method will inevitably introduce additional zeros from $\Delta_y(s)$. But as discussed above, a second-order TIA can avoid these zeros by using an OTA. As well, non-zero τ_3 captures conductance at the OTA's output, leading to more accurate expressions for a practical circuit when compared to [2], [3], [4], [9], [10] which all assume an ideal voltage OpAmp.

At mixer input, the frequency-dependent OTA input impedance Z_{in} is translated to RF when $|\gamma Z_{in}| > R_{sw}$ at small frequency offsets. At large frequency offset, R_{sw} ultimately limits RF filtering to $R_{sw}/(R_{sw} + R_S)$. The benefits of this variable loading on the antenna are implicit in the analysis of the TIA above.

A. No Net Capacitive Positive Feedback ($C_2 \geq 0$)

[8], [16] have shown that the TIA can realize a 2nd-order Butterworth response ($Q = 1/\sqrt{2}$) without employing capacitors C_{2m} and C_{2p} in feedback. In this section, we will develop a simple way to design this type of circuit whose $C_2 \geq 0$.

1) Poles: Consider the equivalent circuit of Fig. 5a. Suppose that $C_2 \geq 0$, then $\tau_2 \geq 0$ and $\tau_z \geq 0$. Using (16b) we now prove with simple inequalities that in the absence of net capacitive positive feedback, Q cannot be arbitrarily large:

$$Q^2 = \frac{k_{in} \times (\tau_1\tau_3 - \tau_2\tau_z)}{(\tau_1+\tau_2+\tau_3)^2} \leq \frac{k_{in} \times (\tau_1\tau_3)}{(\tau_1+\tau_3)^2} \leq \frac{k_{in}}{4}. \tag{19}$$

The relations become equalities when $C_2 = 0$ and $\tau_1 = \tau_3$. $C_2 = 0$ does not necessarily mean absence of capacitive feedback such as in [8]; from (5a), C_2 is also 0 when $C_{2m} = C_{2p}$. This has not been pointed out in the literature so far. Q_{max} is now limited by k_{in}, or R_{in}:

$$Q_{max} = \sqrt{\frac{k_{in}}{4}} \approx \sqrt{\frac{1 + R_S/R_{in}}{4}}. \tag{20}$$

Therefore, when $C_2 = 0$, enforcing $R_{in} = R_S$ for impedance match can lead to Q of $1/\sqrt{2}$.

ω_0 can be calculated from (16a) by setting $\tau_2 = 0$ (since $C_2 = 0$):

$$\omega_0 = \sqrt{\frac{k_{in}}{\tau_1\tau_3}}. \tag{21}$$

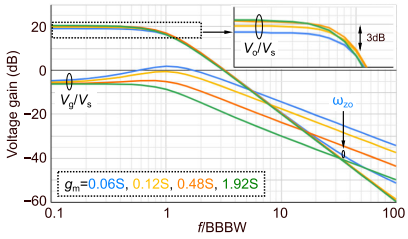


Fig. 6. Simulated TIA transfer functions with Butterworth response ($Q = 1/\sqrt{2}$) following design in (26) with different choices of g_m and corresponding C_F and C_2 ($R_S = 50\Omega$, $A_0 = 21.7$). Unit inverter $g_m = 20\,\text{mS}$ constructed (and used hereafter) with PFET: 175/0.5μm; NFET: 60/0.5μm.

A well-designed 2nd-order system should allow the user to adjust ω_0 and Q independently of each other. This holds true for this circuit. If C_{F1} and C_{F3} are both doubled in value, τ_1 and τ_3 are doubled but ω_0 is halved (21); Q (19) is unchanged. On the other hand, if C_{F1} doubles and C_{F3} is halved, ω_0 is unchanged but Q changes. Thus ω_0 and Q may be tuned independently.

2) Stability: According to (16b), $Q > 0$ when $\tau_2 \geq 0$. Therefore, the poles of the circuit are well-defined and lie in the left-half s-plane. This assures stability.

3) Design Procedure: In a mixer-first receiver, $R_S \approx R_{in}$: (i) $k_{in} = 2$; (ii) $Q = 1/\sqrt{2}$ only if $\tau_1 = \tau_3$ and $C_2 = 0$; (iii) $\omega_0 = \sqrt{2}/\tau_1 = \sqrt{2}/\tau_3$. The TIA can be simply designed and tuned, as follows:

(a) Choose g_m on considerations of power or noise, and to satisfy $g_m \gg \frac{1}{R_S}$, $g_m \gg \frac{A_0}{R_L}$.
(b) Choose $R_F \approx R_S \times A_0$ for impedance match across the wanted channel's bandwidth centered at zero IF.
(c) Adjust capacitance for the Butterworth response, $C_{2m} = C_{2p}$, $C_1 = \frac{\sqrt{2}}{R_S \times \omega_{3\text{-dB}}} - 2C_{2p}$, $C_3 = \frac{\sqrt{2} g_m}{A_0 \times \omega_{3\text{-dB}}} - 2C_{2p}$.

$\omega_{3\text{-dB}} = \omega_0$ is the specified 3-dB bandwidth of the TIA at baseband (BBBW). Since g_m needs only satisfy $g_m R_S \gg 1$, its value is chosen for noise and power. But if g_m is too small, the expressions above may no longer be accurate.

Zeros are calculated using (9):

$$\omega_{zo} = \infty \tag{22a}$$

$$\omega_{zg} = \frac{1}{\tau_3} = \frac{\omega_0}{\sqrt{2}}. \tag{22b}$$

In reality ω_{zo} is affected by the gate-drain capacitance C_{gd} of FETs, which adds to C_{2p}. Depending on the relative size of C_{gd}, this may necessitate use of a larger C_{2m} to neutralize it. But then no degree of freedom remains to adjust $\omega_{eq} = A_0\omega_{zg} = (A_0/\sqrt{2})\omega_{3\text{-dB}} \gg \omega_{3\text{-dB}}$. This is because the constraint $\tau_1 = \tau_3$ to reach $Q = 1/\sqrt{2}$ couples ω_{zg} to ω_0. When A_0 is large enough, the amplifier output port will always limit the overall linearity at blocker frequency. We will determine later the combined effects of 2nd-order rolloff of the filter and the dominant nonlinearity at play that limits intermodulation and gain compression.

B. Using Net Capacitive Positive Feedback ($C_2 < 0$)

Second-order filtering with unconstrained Q can be obtained using one OTA with capacitive positive feedback [4], [5], [7], [10].

To obtain a 2nd-order passband with $Q \geq 1/\sqrt{2}$, according to (19) a mixer-first receiver needs $C_2 < 0$ when $\tau_1 \neq \tau_3$, otherwise a Butterworth response cannot be reached. The positive capacitive feedback must be stronger than the negative, $C_{2m} > C_{2p}$.

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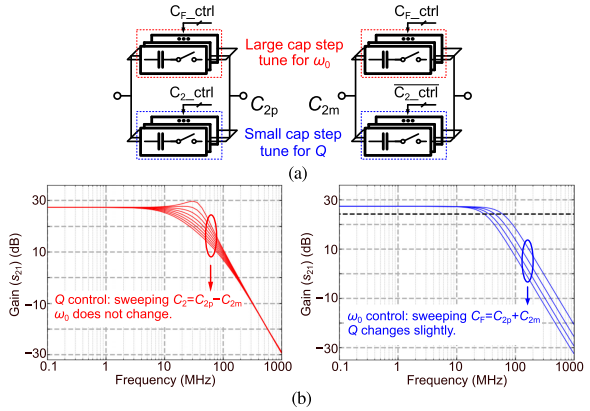


Fig. 7. (a) Implementation of C_{2p} and C_{2m} that independently controls C_F and C_2; (b) Simulated independent ω_0 and Q control through C_F and C_2.

1) Zeros and Poles: ω_0 in (16a) can be simplified further by finding an implicit relation between C_2 and C_{F1}, C_{F3} (defined in Fig. 5c) given (4), and the requirement for stability, $Q > 0 \Leftrightarrow \tau_1 + \tau_2 + \tau_3 > 0$. From definitions (5a) and (5b),

$$|C_2| = |C_{2p} - C_{2m}| < C_F < \min\{C_{F_1}, C_{F3}\}, \tag{23a}$$

$$\text{and } |C_2| = \frac{|\tau_2|}{R_1 R_3 g_m} < \frac{\tau_1 + \tau_3}{R_1 R_3 g_m} = \left(\frac{C_{F1}}{g_m R_3} + \frac{C_{F3}}{g_m R_1}\right) \ll \max\{C_{F1}, C_{F3}\}, \tag{23b}$$

$$\Rightarrow |C_2|^2 \ll C_{F1} C_{F3}, \tag{23c}$$

where $g_m R_3 \approx g_m r_0 \gg 1$, and $g_m R_1 \approx g_m R_S \gg 1$. The only condition to reach (23c) is that $Q > 0$. This means that (23c) is a *universal inequality* for this type of TIA. Using (6)(13)(14)(16a) and (23c), the expression for ω_0 in (16a) is now approximately:

$$\omega_0 \approx \sqrt{\frac{k_{in}}{\tau_1 \tau_3}} = \sqrt{\frac{2}{\tau_1 \tau_3}} = \sqrt{\frac{2}{R_S r_0 C_{F1} C_{F3}}}, \tag{24}$$

where $R_3 \approx r_0$, $R_1 \approx R_S$. (24) is identical to (21). We conclude that ω_0 must be independent of C_2.

Zeros are calculated from (9):

$$\omega_{zo} = \frac{g_m}{C_2} = \underbrace{\sqrt{\frac{(g_m R_S) A_0}{2}}}_{\gg 1} \underbrace{\frac{\sqrt{C_{F1} C_{F3}}}{C_2}}_{\text{Magnitude} \gg 1} \omega_0 \Rightarrow |\omega_{zo}| \gg \omega_0 \tag{25a}$$

$$\omega_{zg} = \frac{1}{\tau_3} = \frac{\omega_0}{\sqrt{2}} \times \sqrt{\frac{\tau_1}{\tau_3}}. \tag{25b}$$

ω_{zo} is located well beyond the filter cut-off frequency, and for practical purposes $\omega_{zo} \to \infty$. Unlike the circuit without net capacitive positive feedback, smaller g_m leads to an increased τ_3, lowering ω_{zg}. If the low-frequency zero causes a peak to appear in the magnitude of the transfer function to the amplifier's input node, this could become the bottleneck to overall linearity. When $\tau_1 = \tau_3$ and $\tau_2 = 0$, (24)(25) become identical to (21)(22). So, the design with $C_2 \geq 0$ in Section III-A is a special case.

Consider when $C_1, C_3 \ll C_F = C_{2p} + C_{2m}$. This holds true when C_1 and C_3 are FET capacitances only and the TIA passband is $\ll$ FET f_T. The amplifier input and output are now loaded equally: $C_{F1} = C_{F3} = C_F$. (23c) becomes $|C_2| \ll C_F$. This simplifies design expressions and tuning method both since τ_1 and τ_3 are tuned by C_F concurrently. From (6)(11)(13)(16)(24)(25), the important design expressions reduce to

$$R_F = A_0 R_S, \tag{26a}$$

$$\omega_0 = \sqrt{\frac{2}{R_S r_0 C_F^2}} = \sqrt{\frac{2 g_m}{A_0 R_S C_F^2}}, \tag{26b}$$

$$\frac{1}{Q} = \frac{(\tau_1 + \tau_3) + \tau_2}{\sqrt{2\tau_1 \tau_3}} = \underbrace{\frac{R_S + r_0}{\sqrt{2 R_S r_0}}}_{\geq \sqrt{2}} + \underbrace{\sqrt{\frac{A_0 (g_m R_S)}{2}}}_{\gg 1} \underbrace{\frac{C_2}{C_F}}_{\leq 0}, \tag{26c}$$

$$\omega_{zg} = \frac{g_m}{A_0 C_F} = \sqrt{\frac{g_m R_S}{2 A_0}} \omega_0, \tag{26d}$$

$$\omega_{eq} = \underbrace{A_0 \omega_{zg}}_{\gg \omega_{zg}} = \frac{g_m}{C_F} = \underbrace{\left(\frac{C_2}{C_F}\right) \omega_{zo}}_{\ll |\omega_{zo}|} = \sqrt{\frac{A_0 (g_m R_S)}{2}} \omega_0. \tag{26e}$$

Q^{-1} in (26c) consists of two terms. The first term is determined only by OTA design. To reach Butterworth response, this requires the second term to be ≤ 0. When $R_S = r_0$ (i.e. $\tau_1 = \tau_3$), $Q = 1/\sqrt{2}$ requires $C_2 = 0$ (i.e. $\tau_2 = 0$). When $R_S \neq r_0$, $Q = 1/\sqrt{2}$ requires $C_2 < 0$, as expected. ω_{eq} is calculated from (11) and $|\omega_{zo}| \gg \omega_{eq} \gg \omega_{zg}$.

Using (26), we now illustrate the design of a 2nd-order TIA. Simulation of an inverter amplifier with ideal RC elements yields the frequency response of Fig. 6. Here $A_0 = 21.7$, $R_S = 50\Omega$, BBBW$= 4$ MHz, and for a given g_m (26) specifies all capacitances and resistances. The design equations are accurate over a wide range of g_m, even when $g_m R_S$ is as low as 3. A higher g_m pushes out ω_{zg}, ω_{eq} and reduces gain peaking in V_g/V_s at the input port. ω_{zo} is located at a high frequency $f_{zo} \gg 10 \times$ BBBW.

Q of this filter is much more sensitive to C_2 than to C_F, because (23c) and (26c) tell us that $|C_2| \ll C_F$:

$$\left|\frac{\partial Q^{-1}}{\partial C_F}\right|_{C_2} = \left|\frac{\partial Q^{-1}}{\partial C_2}\right|_{C_F} \times \frac{|C_2|}{C_F} \ll \left|\frac{\partial Q^{-1}}{\partial C_2}\right|_{C_F}. \tag{27}$$

Although C_2 is insignificant and ignored in (24), it cannot be ignored in (26c) because it is weighted by a large coefficient of opposite sign. Even when $C_{2m} \approx C_{2p}$, a tiny difference between C_{2p} and C_{2m} can affect Q. Therefore, C_2 must be tuned at a finer resolution than C_F. Fig. 7a shows capacitor banks that control C_F and C_2 independently. As a result, effectively $C_{2p} = (C_F + C_2)/2$ and $C_{2m} = (C_F - C_2)/2$. This arrangement simplifies design and tuning by decoupling parameters for ω_0 and Q. Previous works [7], [10] could have benefited from this method of tuning. The simulations in Fig. 7b (inverter $g_m = 320$ mS) show an example of the independent control of ω_0 and Q by C_F and C_2. $Q = 1/\sqrt{2}$ gives a maximally flat lowpass characteristic, when ω_0 is also $\omega_{3\text{-dB}}$.

When $|C_2|$ is "insignificant" as in (26c), $|\omega_{zo}| = |g_m/C_2|$ lies at a very high frequency that may approach FET f_T. For large FET sizes, the gate-drain capacitance C_{gd} adds to C_{2p} and raises C_2. As a result, Q is smaller compared to its expected value when FET gate-drain capacitance is neglected. This leads to lower gain at ω_0 because $|V_o/V_s(j\omega_0)| = Q\,|V_o/V_s(0)|$, and to a lower 3-dB BW. It is seen in Fig. 6 (C_{gd} of FET was ignored in analysis) where larger g_m leads to lower gain at ω_0. Therefore, designers should include C_{gd} into their calculation, or neutralize it as in [18].

2) Stability: The TIA is designed for a nominal source impedance of $R_S = 50\,\Omega$, and its well-defined poles show that it is stable. But a realistic antenna impedance is modeled as $R_{S,r} \parallel jX_{S,r}$: this is not constant, but varies over some range of VSWR [19]. By definition, $1/\text{VSWR} < R_{S,r}/R_S < \text{VSWR}$.

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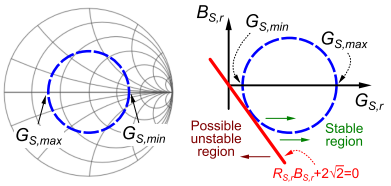


Fig. 8. The TIA source admittance confined within a VSWR circle (left) in the admittance plane (right) must locate entirely in the stable region specified by (31) to guarantee stability. ($G_{S,r} = 1/R_{S,r}$).

$X_{S,r}$ is roughly constant over the BB bandwidth when the antenna impedance varies slowly at RF, i.e., when $Q_{ANT} \ll f_{LO}/(2\text{BBBW})$ [19].

First we consider stability with a resistive antenna. If $B_{S,r} = 1/X_{S,r} = 0$, then when the real part of source impedance R_S changes with VSWR to some $R_{S,r}$, Q in (16b) also changes:

$$Q|_{R_{S,r}} = \left.\frac{\sqrt{k_{in}(\tau_1\tau_3)}}{\tau_1+\tau_2+\tau_3}\right|_{R_{S,r}} \approx \left.\frac{\sqrt{k_{in}(\tau_1\tau_3)}}{(\tau_1+\tau_2)|_{R_S}\times\frac{R_{S,r}}{R_S}+\tau_3}\right|_{R_{S,r}} \tag{28}$$

$$\Rightarrow\ Q|_{R_S} = \frac{1}{\sqrt{2}} \text{ when } (\tau_1+\tau_2)|_{R_S} = \left.\left(\sqrt{2\sqrt{2}\tau_1\tau_3}-\tau_3\right)\right|_{R_S}, \tag{29}$$

where $\tau_1 \propto R_1 \approx R_{S,r}$ and from (14) $k_{in} \approx 2$. The approximation in (28) is valid when VSWR $\ll A_0$, which is always true in practice. If $(\tau_1+\tau_2)|_{R_S} \geq 0$, then $Q|_{R_{S,r}} > 0$; this guarantees unconditional stability. But it requires $\tau_3|_{R_S} \leq 2\sqrt{2}\,\tau_1|_{R_S}$. If $C_1 = C_3 = 0$, then $r_0 \leq 2\sqrt{2}R_S$. However, when $r_0 > 2\sqrt{2}R_S$ (e.g., small g_m), then for unconditional stability C_1 must be larger; that is:

$$C_{F1} \geq \frac{r_0}{2\sqrt{2}R_S} \times C_{F3}. \tag{30}$$

Interestingly, this imposes the upper bound on $\tau_3 \leq \sqrt{4\sqrt{2}}/\omega_0$ and therefore, from (25b), a lower bound on $\omega_{zg} \geq \omega_0/\sqrt{4\sqrt{2}}$ or on ω_{eq} $(= A_0\omega_{zg})$. Thus when ω_{zg} or ω_{eq} are located at a high enough frequency to improve blocker tolerance, stability is also guaranteed.

Next, we extend the result of (30) to examine whether stability can be guaranteed when antenna impedance is complex. Negative (inductive) antenna susceptance $B_{S,r}$ in parallel with C_{F1} can lower τ_1 so much that $(\tau_1+\tau_2) < 0$. Then from (28) if the ratio $R_{S,r}/R_S$ is large enough, Q will become <0. This indicates an unstable circuit. The constant $B_{S,r}$ can be replaced by an equivalent capacitance $C_{S,r} = B_{S,r}/\omega_{ph,T}$, where $\omega_{ph,T}$ is the frequency where $|\angle T(j\omega_{ph,T})| = 180°$. The substitution does not alter the gain margin, nor monotonicity[4] of $|\angle T(j\omega)|$, so stability is identical with and without replacing $B_{S,r}$. When $r_0 = 2\sqrt{2}R_S$, $\tau_1 = |\tau_2| = \tau_3/2\sqrt{2}$. So $\omega_{ph,T} \approx 1/\tau_1$, where $Z_1(j\omega)$, $Z_3(j\omega)$ and y_{12} each contribute approximately 45° (τ_1), 90° (τ_3) and 45° (τ_2) to $|\angle T(j\omega_{ph,T})| = 180°$, respectively. Therefore, $C_{S,r} \approx \tau_1 B_{S,r}$. Stability requires $\tau_1+\tau_2+\tau_3 > 0$. Expressed in terms of circuit elements, this means

$$\left(R_S(C_{F1}+C_{S,r})+\tau_2\right)\Big|_{R_S} \times \frac{R_{S,r}}{R_S} + \tau_3 \geq (R_{S,r}B_{S,r}+2\sqrt{2}) \times \tau_1|_{R_S,B_{S,r}=0} > 0. \tag{31}$$

This becomes an equality when $C_{F1} = (r_0/(2\sqrt{2}R_S))C_{F3}$. As shown in Fig. 8, the inequality (31) defines the stable region in the admittance plane.

For a given antenna VSWR, $R_{S,r}$ and $B_{S,r}$ cannot be arbitrary, but are confined to a circle in the admittance plane[5] shown in Fig. 8. A large circle (large VSWR) can cross the stability boundary. It can be shown that (31) is satisfied when VSWR< 5.8: this means that the circuit is stable for this range of VSWR when, referring to (30), $C_{F1} = (r_0/(2\sqrt{2}R_S))C_{F3}$. This covers typical variations in antenna impedance. If VSWR might be higher, C_{F1} should be larger to expand the stable region in Fig. 8.

3) Design Procedure: g_m cannot be scaled down indefinitely to save power, because: (i) its input-referred voltage noise rises; (ii) design expressions are less accurate; and (iii) ω_0 and Q can no longer be tuned independently.

In nominal conditions, $g_m R_S \gg 1$ requires that $g_m \gg 20\,\text{mS}$. With small g_m and therefore larger τ_3, the zero ω_{zg} in the transfer function to the amplifier input may lie at a low frequency in the filter passband, producing a peak close to a blocker's frequency. To prevent this, ω_{eq} must be placed at some high frequency. We will discuss this at greater length in Section IV. Instead of scaling up g_m to raise τ_1, a larger C_1 may be used. Then $\tau_3 = 2/(\omega_0^2\tau_1)$. ω_0 and Q can still be tuned independently.

In summary, the TIA is designed and tuned as follows:

(a) Choose g_m to meet specifications on power and noise. Check that it satisfies (18) $g_m \gg \frac{1}{R_S}$, $g_m \gg \frac{A_0}{R_L}$.
(b) Choose $R_F \approx R_S \times A_0$ for impedance match.
(c) Adjust C_F, C_1 and C_3 to tune $\omega_0 = \omega_{3\text{-dB}}$ (24). Eliminating C_1 and C_3 simplifies design (26), but this is not feasible for small g_m on considerations of stability (30) and blocker tolerance (25b).
(d) Adjust C_2 (26c) for the Butterworth response $Q = 1/\sqrt{2}$.

Not surprisingly, this design process is similar to that in Section III-A. They differ in tuning method for Butterworth response because Sec. III-A presupposes that $\tau_1 = \tau_3$.

C. Choices of OTA Circuit

The analysis so far assumes a constant g_m that does not change with frequency. For an inverter amplifier [20], this is almost true. Usually it is also true of single-stage OpAmps (e.g., cascode amplifier), whose internal nodes are low-impedance. But in single-stage OpAmps power consumption scales with g_m.

For the same power consumed, two-stage OTAs can yield higher voltage gain A_0, higher transconductance g_m, and lower output impedance r_0. For example, a source follower as the output stage boosts $g_m = A_0/r_0$ (by lowering r_0) without adding substantial power. However, multiple stages will add to the number of poles. If the dominant pole of the internal nodes is $\omega_{int} = 1/\tau_{int}$, then g_m in Fig. 2a may be replaced by $g_m(s) = g_{m0}/(1+s\tau_{int}) \equiv g_{m0}/\Delta_{g_m}(s)$. (3) now becomes

$$\frac{V_o}{V_s}(s) \propto \frac{(y_{21,N}\Delta_{g_m}+g_{m0}\Delta_y)\Delta_y}{y_{11,N}y_{22,N}\Delta_{g_m}-y_{12,N}(y_{21,N}\Delta_{g_m}+g_{m0}\Delta_y)}(s) \tag{32a}$$

$$\frac{V_g}{V_s}(s) \propto \frac{y_{22,N}\Delta_y\Delta_{g_m}}{y_{11,N}y_{22,N}\Delta_{g_m}-y_{12,N}(y_{21,N}\Delta_{g_m}+g_{m0}\Delta_y)}(s). \tag{32b}$$

[4]Phases of Z_1, Z_3 and y_{21} all increase with frequency and the sign of the slope is unchanged.

[5]The Möbius transformation (e.g., the Smith Chart) maps every generalized circle (a line or a circle) to a generalized circle. Therefore, a VSWR circle on the Smith Chart maps to a circle on the admittance plane.

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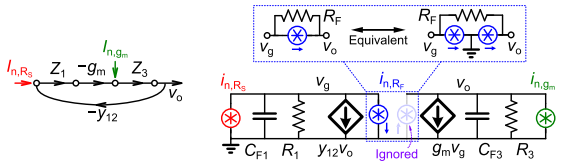


Fig. 9. TIA signal flow graph and equivalent circuit for noise analysis. Blue arrows reveal noise correlation of two R_F noise sources, one of which is later ignored in analysis.

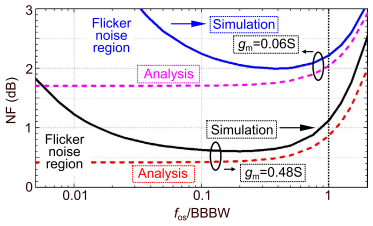


Fig. 10. TIA noise figure simulation versus analysis. (BBBW= 40MHz).

$V_o/V_s(s)$ has two zeros from $g_m = -y_{21}$, but as before they are likely to be at high frequencies ($|\omega_{zo}| > \sqrt{g_m/(C_2\tau_{int})} \gg \tau_{int}^{-1}$). $V_g/V_s(s)$ has one more zero from $1/\tau_{int}$.

A source follower [8] can push $1/\tau_3$ to high frequency, and sets τ_{int} to equal the original τ_3. The polynomial $\Delta_{TF}(s)$ is now:

$$\Delta_{TF}(s) \approx \big(1 + T(s)\big) \times (1 + s\tau_1)(1 + s\tau_{int}). \tag{33}$$

Therefore, when evaluating the frequencies of the pole and zero with (9b)(16), τ_{int} replaces τ_3. The OTA's internal pole must be carefully chosen, but in other respects the design flow remains the same.

Using a high-speed OpAmp [10] pushes both $1/\tau_{int}$ and $1/\tau_3$ to high frequency. But then the circuit cannot be designed with positive-valued components as discussed in Section II, unless a different feedback network is used. Doing so will result in $\Delta_y(s)$ introducing low-frequency zeros into $V_o/V_s(s)$ [10].

D. TIA Noise Figure and Choice of g_m

The noise factor $F(\omega)$ of the TIA in Fig. 5a using an inverter amplifier is calculated from the noise currents shown in Fig. 9. The noise current of the input source is i_{n,R_S}. The FET noise current i_{n,g_m} appears at the TIA output. Using the source shifting theorem, the R_F noise current i_{n,R_F} is split into two identical current sources that appear across the TIA's input and output ports (Fig. 9), respectively. All noise current sources now connect across the TIA input or output port. The output noise transfer functions of i_{n,R_S} and i_{n,g_m} are

$$H_1(s) = \frac{V_o}{I_{n,R_S}}(s) = \frac{-Z_1(s)g_m Z_3(s)}{1 + T(s)} \tag{34a}$$

$$H_2(s) = \frac{V_o}{I_{n,g_m}}(s) = \frac{Z_3(s)}{1+T(s)} = \frac{-(1+s\tau_1)}{g_m R_S} \times H_1(s). \tag{34b}$$

Because $g_m R_S/|1 + j\omega\tau_1| \gg 1$ from (18), the R_F noise current at the TIA output may be ignored (Fig. 9). Therefore, $V_o/I_{n,R_F}(s) \approx V_o/I_{n,R_S}(s)$. The noise factor can then be calculated from (34):

$$F(\omega) \approx \frac{|H_1(j\omega)|^2(\frac{1}{R_S} + \frac{1}{R_F}) + |H_2(j\omega)|^2\gamma_n g_m}{|H_1(j\omega)|^2 \times \frac{1}{R_S}}$$

$$= 1 + \frac{\gamma_n}{g_m R_S}[1 + (\omega\tau_1)^2] + \frac{R_S}{R_F}, \tag{35}$$

where $\gamma_n \approx 1.3$ is the FET thermal noise coefficient. Fig. 10 shows NF simulation versus analysis results using circuits in Fig. 6, but with BBBW= 40 MHz. (35) predicts well the NF due to white noise. Flicker noise, not modelled by (35), can be lowered if necessary using FETs with larger gate area.

From (35), NF rises at channel edge ω_0 as shown in Fig. 10. τ_1 should be small to minimize the rise. However, since $\tau_3 = 2/(\omega_0^2\tau_1)$, this can as a consequence hurt both linearity (9b)(11) and stability (30). This trade-off between noise and linearity can be mitigated at the cost of higher power (large $g_m R_S$). This should be taken into account when choosing g_m during design in Sections III-A and III-B. Too high an ω_{eq} or ω_{zg} for better linearity means a larger power dissipation than is necessary to meet specifications on noise.

TABLE I
SUMMARY OF NOTATIONS FOR DISTORTION ANALYSIS

Notation	Definition
g_{ij}	3rd-order distortion coefficients of FET drain current
V_{A0}	Early voltage at bias point
V_{D0}	Drain voltage such that extrapolated $V_A = 0$
k_{VS}	Velocity saturation factor
$V_{d,IP3,g_{ij}}$, $V_{IIP3,g_{ij}}$	OIP3 voltage from g_{ij} for blocker case 1
$V_{IIP3,g_{ij}}$, $V_{IIP3,2,g_{ij}}$	IIP3 voltage from g_{ij} for blocker case 1, 2
$\omega_{IP3,i}$, $\omega_{IP3,2,i}$	Frequency of equal contribution to IP3 for blocker case 1, 2
$V_{d,B1dB}$	Drain-limited drain-referred B1dB
$V_{g,B1dB,g_{30}}$	Gate-limited gate-referred B1dB
ω_{B1dB}	Frequency of equal contribution to B1dB

IV. DISTORTION FROM OOB BLOCKERS

At frequency ω_{eq} the sinewave amplitude at the TIA's input and output ports is equal (Fig. 4a). This is not, however, the lower distortion intercept frequency that is shown in Fig. 4b. Distortion originates at the FET gate from a different mechanism than at its drain. The following analysis finds expressions for OOB-IIP3 and B1dB that characterize distortion in an inverter amplifier as a function of the blocker offset frequency ω_{os}.

To explain our approach in what follows, we remind the reader that IP3 is found by extrapolation of 3rd-order nonlinearity induced by *small signals*. In the following section we derive the distortion coefficients and resulting IP3. In contrast, the 1-dB compression point is determined by a large sinewave input that, due to the gradual onset of nonlinearity, induces a drop in effective gain to the fundamental frequency in the output response. In circuits with weak nonlinearity, this is also predicted by the coefficients of small-signal distortion. But when the nonlinearity is strong so that compression sets in abruptly, it calls for a full-fledged *large-signal* analysis. This is what we employ in the second section below.

Table I summarizes the notations used in this section.

A. OOB-IIP3

Following (8), at OOB frequencies when $\omega_{zo} \gg \omega \gg 1/\tau_1, 1/\tau_3$ the transfer functions are:

$$\frac{V_o}{V_s}(s) \approx \frac{A_{vo}}{s^2/\omega_0^2} = \frac{1}{s\tau_1} \times (-A_0) \times \frac{1}{s\tau_3} \tag{36a}$$

$$\frac{V_g}{V_s}(s) \approx \frac{A_{vg}(s/\omega_{zg})}{s^2/\omega_0^2} = \frac{1}{s\tau_1}. \tag{36b}$$

The equivalent circuit of Fig. 11 represents (36a) and (36b). The approximations used in these expressions are consistent with $|T(j\omega)| \ll 1$ in (7) at frequencies when $1/\tau_z \gg \omega \gg 1/\tau_1, 1/\tau_3$; that is, when at OOB frequencies the loop gain

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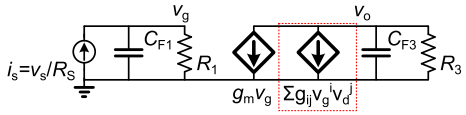


Fig. 11. TIA equivalent circuit for distortion analysis.

has dropped below 1 so that the TIA is, in effect, operating open loop.

The OTA small-signal output current $i_d(v_g, v_d)$ (body and source connected to fixed voltages) at bias voltage (V_G, V_D) including 3rd-order distortion can be written as:

$$i_d(v_g, v_d) = \underbrace{g_m v_g + g_d v_d}_{\text{Linear terms}} + \sum_{i=3-j=0}^{3} g_{ij} v_g^i v_d^j, \tag{37}$$

where g_m is the transconductance, $g_d = 1/r_0$ is the output conductance, and g_{ij} are the *3rd-order distortion coefficients*. The four coefficients specify the joint effect of two independent distortion mechanisms in the output i_d, one due to v_g and the other due to v_d. The equivalent circuit of Fig. 11 models the distortion currents with the nonlinear voltage-controlled current source. The coefficients can be calculated from the expression for large-signal drain current $I_d(V_g, V_d)$. In an NFET biased in strong inversion with body and source grounded, the drain current follows the square law:

$$I_d \approx \underbrace{\tfrac{1}{2}\beta_{eff} V_P^2}_{\text{function of } V_g} \times \underbrace{\left(1 + \int_{V_{dsat}}^{V_d} V_A^{-1} dv\right)}_{\text{function of } V_d} \tag{38}$$

and $\beta_{eff} = \frac{\beta}{1+\theta V_P}$ [21] accounts for velocity saturation, $\theta = 1/(E_c L)$, $E_c^{-1} \simeq 0.2\,\mu$m/V, the pinchoff voltage $V_P = (V_g - V_{tn})$, V_A is the Early voltage, V_{dsat} is the drain saturation voltage.[6] To include higher-order distortions caused by drain voltage, a V_d-dependent Early voltage $V_A(V_d)$ must be used. BSIM3v3 [22] describes V_A as a linear function of V_d:

$$\begin{aligned} V_A(V_d) &= V_{Asat} + \frac{V_d - V_{dsat}}{V_D - V_{dsat}} \times (V_{A0} - V_{Asat}) \\ &\equiv \frac{v_d + V_{D0}}{V_{D0}} \times V_{A0}, \end{aligned} \tag{39}$$

where V_{Asat} is the Early voltage at $V_d = V_{dsat}$ when the FET enters the saturation region, $V_{A0} = V_A(V_D) = I_D r_0$ is the Early voltage at DC bias $V_d = V_D$, V_{D0} is a constant voltage introduced for simplicity, such that $V_A = 0$ at $V_d = V_D - V_{D0}$. (39) considers only channel length modulation (CLM). We ignore drain-induced barrier lowering (DIBL), which will dominate output conductance at higher V_d [23].

(38) and (39) are sufficient to find all coefficients g_{ij} in (37):

$$\text{Drain only: } g_{03} = \left.\frac{\partial^3 I_d}{6\,\partial V_d^3}\right|_{V_G,V_D} = \frac{g_d}{3V_{D0}^2}, \tag{40a}$$

$$g_{12} = \left.\frac{\partial^3 I_d}{2\,\partial V_g \partial V_d^2}\right|_{V_G,V_D} = -\frac{g_m}{2V_{A0}V_{D0}}, \tag{40b}$$

$$g_{21} = \left.\frac{\partial^3 I_d}{2\,\partial V_g^2 \partial V_d}\right|_{V_G,V_D} = \frac{\beta_{eff}}{2V_{A0}k_{VS}^2}, \tag{40c}$$

$$\text{Gate only: } g_{30} = \left.\frac{\partial^3 I_d}{6\,\partial V_g^3}\right|_{V_G,V_D} = -\frac{\beta_{eff}\theta}{2k_{VS}^3}, \tag{40d}$$

where $V_{P0} = V_P(V_G)$ at DC bias, $k_{VS} = 1 + \theta V_{P0}$. In an inverter amplifier, these parameters represent the composite nonlinearity arising from NFET and PFET. g_{30} is recognizable as the 3rd-order g_m nonlinearity coefficient.

(37) is now evaluated by applying KCL separately at the fundamental frequency and the IM3 frequency:

$$\text{Fundamental: } \Big(g_m v_g + \big(g_L(s) + g_d\big)v_d\Big) = 0, \tag{41a}$$

$$\text{IM3: } \Big(\big(g_L(s) + g_d\big)v_d + \sum_{i=3-j=0}^{3} g_{ij} v_g^i v_d^j\Big) = 0, \tag{41b}$$

where $g_L(s) = 1/R_L + sC_{F3}$ is the load admittance. To the first order, IM3 tones generated by $\sum g_{ij} v_g^i v_d^j$, are calculated assuming v_g and v_d contain fundamental frequencies only. Our calculations assume that IM3 is produced by inputs well below compression. With these simplifications, (41a) decouples from (41b).

1) Adjacent $f_{blk1,BB}$ and $f_{blk2,BB}$: In this first scenario the two blockers are adjacent in frequency and generate adjacent IM3 tones ($f_{blk1,BB} - f_{blk2,BB} \ll$ BBBW). They are not necessarily close-in blockers. Although IM3 tones are not in-band, this scenario paves the way to understand in-band IM3 blocker scenario discussed later. The TIA filters all these frequencies nearly equally. OIP3 is calculated simply by finding the fundamental and 3rd-order coefficients of v_d from (41). Each g_{ij} gives an OIP3 according to (37) and (41):

$$V_{d,IP3,g_{03}} = 2V_{D0}\sqrt{\frac{g_L + g_d}{g_d}} \approx 2V_{D0}\sqrt{|1 + j\omega_{os}\tau_3|} \tag{42a}$$

$$V_{d,IP3,g_{12}} \approx V_{d,IP3,g_{03}} \times \sqrt{\frac{A_{gd}V_{P0}}{3V_{D0}}} \approx \sqrt{\frac{4A_0 V_{P0} V_{D0}}{3}} \tag{42b}$$

$$V_{d,IP3,g_{21}} \approx V_{d,IP3,g_{12}} \times \sqrt{\frac{A_{gd}V_{P0}}{V_{D0}}} \tag{42c}$$

$$V_{d,IP3,g_{30}} \approx V_{d,IP3,g_{21}} \times \sqrt{\frac{2A_{gd}}{A_0\theta V_{P0}}}, \tag{42d}$$

where $A_{gd}(j\omega) = |g_m/(g_L(j\omega) + g_d)| \approx |A_0/(j\omega\tau_3)|$ is the magnitude of the small-signal voltage gain from gate to drain. The approximations above assume $V_{A0} = I_D r_0 \approx 0.5 g_m V_{P0} r_0 = 0.5 A_0 V_{P0}$; $k_{VS} \approx 1$; and $g_L + g_d \approx g_L$ when $\omega\tau_3 \gg 1$. Referred to the input:

$$V_{IIP3,g_{ij}}(j\omega) = \frac{V_{d,IP3,g_{ij}}(j\omega)}{|V_o/V_s(j\omega)|}. \tag{43}$$

If $g_L + g_d \approx g_L \approx \omega C_{F3}$, each OIP3 term $V_{d,IP3,g_{ij}}$ in (42) defines an asymptotic straight line in the OIP3-ω_{os} plot. Fig. 12a shows an example and a variation, using the same circuit as in Fig. 6. Table II contains the extracted parameters for FETs of channel length $L = 0.5\,\mu$m used in the simulation. At low ω_{os}, g_{03} and large drain voltage dominates distortion. At frequencies above $1/\tau_3$, OIP3 rises at the rate of 10 dB/dec; see (42a). When $\omega_{os} > \omega_{IP3,1}$, OIP3 becomes constant with frequency because g_{12} determines distortion; see (42b). Then, when $\omega_{os} > \omega_{IP3,2}$ g_{21} will be dominant, causing OIP3 to fall at a rate of 10 dB/dec; see (42c). At somewhat higher frequency when $\omega_{os} > \omega_{IP3,3}$, g_{30} prevails: now gate voltage limits OIP3, which drops with

[6] V_{dsat} and V_P are different when short-channel effects are prominent, such as when carriers reach their drift saturation velocity.

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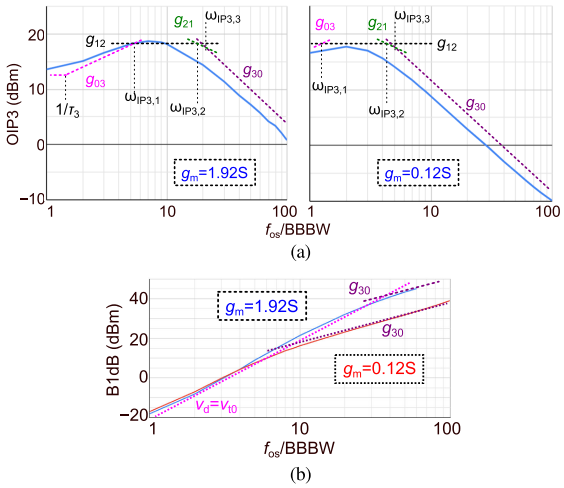


Fig. 12. TIA (a) OOB-OIP3 (adjacent $f_{blk1,BB}$ and $f_{blk2,BB}$) simulation versus analysis and (b) OOB-B1dB simulation of the same circuit versus analysis. (Simulation in solid curves; Analysis in dotted lines).

TABLE II

VALUES OF PARAMETERS USED IN ANALYSIS

V_{P0}	V_{D0}	A_0	θ
0.35 V	0.67 V	21.7	0.4 V^{-1}

frequency at 20 dB/dec; see (42d). The corner frequencies defining these regimes calculated from (11) and (42) are:

$$\omega_{IP3,1} = \frac{A_0 V_{P0}}{3\tau_3 V_{D0}} = \frac{V_{P0}}{3V_{D0}} \times \omega_{eq} \tag{44a}$$

$$\omega_{IP3,2} = 3\omega_{IP3,1} = \frac{V_{P0}}{V_{D0}} \times \omega_{eq} \tag{44b}$$

$$\omega_{IP3,3} = \frac{2}{A_0\theta V_{P0}} \times \omega_{eq}. \tag{44c}$$

These corner frequencies stay the same when distortion is referred to the input. The text leading up to (44) indicates that through g_{12} and g_{21}, the gate voltage plays a role on overall IP3 even when its voltage swing is much less than the swing of the drain voltage. Short of this analysis, this point would not be obvious.

Since $V_{d,IP3,g_{12}}$ in (42b) is independent of frequency, it sets the upper bound on OIP3 (Fig. 12a). To maximize OIP3 for nearby blockers, $\omega_{IP3,1}$ should be retracted to as low a frequency as is feasible. The peak OIP3 will remain constant until the frequency $\omega_{IP3,2}$, which (44b) shows is 3× higher—provided that $\omega_{IP3,3} > \omega_{IP3,2}$. At best, then, peak OIP3 is achieved over the frequency band $(\omega_{IP3,1}, 3\omega_{IP3,1})$. ω_{eq} should be chosen so that the largest nearby blockers land in the frequency range of maximum OIP3. This leads to an optimum g_m when, according to (26), $C_1 = C_2$. If $C_1 \neq C_2$ (e.g., design of Section III-A) the freedom to choose g_m remains.

$\omega_{IP3,3}$ must be $\geq\omega_{IP3,2}$ to realize the full frequency band over which OIP3 is at its peak. Carrier velocity is lower at longer channel length since $\theta \propto L^{-1}$. Invoking (44b) and (44c), if the pinchoff voltage at DC bias V_{P0} is lowered so that $V_{P0} \leq V_{P23} \triangleq \sqrt{(2V_{D0})/(A_0\theta)}$, then $\omega_{IP3,3}$ will be $\geq \omega_{IP3,2}$. In our case, $V_{P23} \approx 0.38V \approx V_{P0}$. As a result, $\omega_{IP3,2} \approx \omega_{IP3,3}$ as is shown in Fig. 12a.

2) IB IM3 When $f_{blk2,BB} - 2f_{blk1,BB} \ll$ BBBW: In this second scenario, the two blockers might be far apart in frequency but they generate IM3 in the TIA passband. A commonly used test condition to measure IP3 is to apply two input tones, one

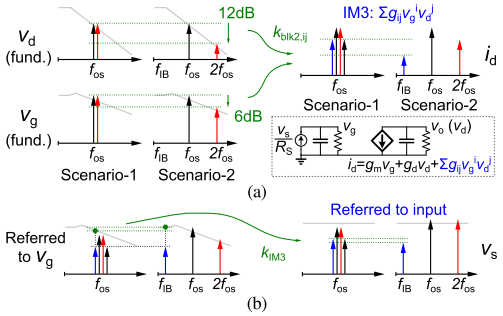


Fig. 13. Two blocker scenarios comparison on input-referred IM3 because of filtering effect on (a) blockers fundamental tones, and (b) IM3 tone.

TABLE III

SIMULATION VS. ANALYSIS OF OOB-IIP3 (IB IM3)

	JSSC'18 [7]	This work (g_m = 320 mS)
f_{os} range (BBBW)	2-4	2-4
Sim. $\lvert V_o/V_s \rvert$ Slope (dB/dec)	36.5†	40
Sim. OOB-IIP3 Slope (dB/dec)	46.5†	46
Calc. OOB-IIP3 Slope (dB/dec)	46.5 (from g_{12})	50 (from g_{12})

†: estimated from figures.

at $f_{blk1,BB} = f_{os}$ and the other at $f_{blk2,BB} = 2f_{os} \pm f_{IB}$, where $f_{IB} \ll$ BBBW. The IM3 tone is at f_{IB}, which is always in-band whatever f_{os}.

Here the IP3s will be different from what (42), (43) predict, for two reasons: (i) greater filtering at the input frequency $f_{blk2,BB}$ than at $f_{blk1,BB}$; and (ii) unfiltered transmission of the IM3 frequency f_{IB}. This is shown in Fig. 13a and 13b. Since $f_{blk2,BB} \approx 2f_{os}$, then relative to $f_{blk1,BB}$ it is filtered by −6 dB and −12 dB, respectively, at the input and output of the TIA. The additional filtering at $f_{blk2,BB}$ lowers the IM3 drain current $g_{ij} v_g^i v_d^j$, respectively, by factors of $k_{blk2,ij}$ =4, 3, 2.4, 2 for g_{03}, g_{12}, g_{21}, g_{30}. This increases V_{IIP3} by factor of $\sqrt{k_{blk2,ij}}$. Also, the IM3 frequency f_{IB} lies in the passband and experiences higher voltage gain, which lowers input-referred IM3 by another factor of $k_{IM3} \triangleq \big(V_g/V_s(0)\big)/\big(V_g/V_s(\omega_{os})\big) = \omega_{os}\tau_1/2$. This raises V_{IIP3} by another $\sqrt{k_{IM3}}$. Altogether the 3rd-order intercept voltage $V_{IIP3,2,g_{ij}}$ in this second scenario is related to $V_{IIP3,g_{ij}}$ in the first scenario of adjacent blockers by:

$$V_{IIP3,2,g_{ij}} = V_{IIP3,g_{ij}} \times \sqrt{k_{blk2,ij} k_{IM3}}, \quad i, j \in [\![0, 3]\!]. \tag{45}$$

Since $k_{IM3} \propto \tau_1/2 = \omega_{eq}/(A_0\omega_0^2)$, therefore, a higher ω_{eq} should improve IIP3. k_{IM3} also introduces in $V_{IIP3,2,g_{ij}}$ an additional 10 dB/dec rolloff vs. frequency. At the frequencies where g_{12} dominates in 2nd-order TIAs, this translates to a cumulative rolloff vs. frequency at 40+10=50 dB/dec. This steep lowering of distortion, or rise in OIP3 is observed in simulation, reported by [7] as shown in Table III, and confirmed by our own measurements (Fig. 18b).

Next we analyze the critical frequencies $\omega_{IP3,2,i}$ where two IP3s $V_{IIP3,2,g_{ij}}$ are equal, and how they relate to $\omega_{IP3,i}$ for this second blocker scenario. k_{IM3} does not change the critical frequency, as it is independent of the index ij. But $k_{blk2,ij}$ varies with index ij. It changes the critical frequencies $\omega_{IP3,i}$ in (44) to

$$\omega_{IP3,2,i} = \frac{k_{blk2,[i-1][4-i]}}{k_{blk2,[i][3-i]}} \omega_{IP3,i} \approx 0.8\omega_{IP3,i}, \quad i \in [\![1, 3]\!], \tag{46}$$

where the error in the approximation is <0.5 dB. (46) implies that all intercept frequencies are lowered by 0.8× compared to (44), and maintain their original order. The design conclusions of Section IV-A.1 remain valid.

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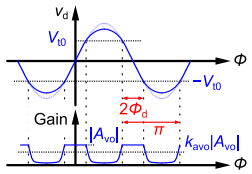


Fig. 14. Small-signal gain reduces (from $|A_{vo}|$ to $k_{avo}|A_{vo}|$) when FETs are at triode region with a strong blocker.

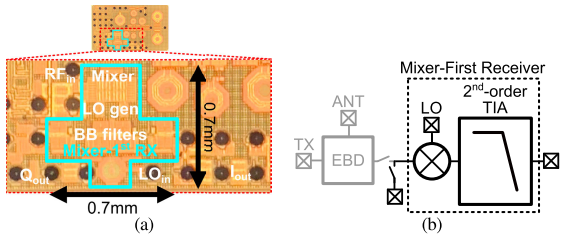


Fig. 15. (a) Die micrograph of the mixer-first receiver fabricated in 65-nm RF-SOI CMOS process (b) as part of a monolithic duplexer.

We summarize the optimum design strategy for large IP3. At the optimum, (a) the region determined by g_{03} is shrunk to the extent possible by lowering $\omega_{IP3,2,1}$; (b) $\omega_{IP3,2,2}$ and $\omega_{IP3,2,3}$ are pushed out in frequency to where the inputs have been filtered and do not distort at baseband, but where distortion in the mixer arising from the unfiltered inputs remains. This allows the TIA to yield the highest OOB-IIP3 at low offset frequencies determined by g_{12}. The intercept point rises with frequency offset at the rate of 50 dB/dec.

B. OOB Blocker-Induced 1-dB Gain Compression (B1dB)

For weakly nonlinear circuits, it is well known that the IIP3 is ∼10 dB higher than the 1-dB compression point (P1dB). There is another 3 dB difference between P1dB and B1dB. So for these circuits, B1dB lies ∼13 dB below IIP3.

But a large input might force the FET out of its saturation region into triode. The resulting nonlinearity in drain current i_d at the FET's *output port* is no longer weak in the mathematical sense, but becomes drastic. The output resistance drops dramatically as the FET is driven deeper into its triode region, causing a collapse of the small-signal gain in that region (Fig. 14). In practice, this strong nonlinearity at the TIA output will determine the B1dB.

When a FET is driven from saturation into triode, its output resistance changes by the ratio $\approx r_0/(2/g_m) = A_0/2 \gg 1$. To a rough approximation, we can say that the small-signal voltage gain in triode drops to 0. This approximation has worked satisfactorily in [24] to predict RF distortion in single-FET large-signal amplifiers.

Assume that the linearly amplified output voltage due to a blocker input is $v_{d,nclp} = V_{d0}\sin(\omega_{blk}t)$, where ω_{blk} is the translated blocker frequency at baseband. When $\omega_{blk} < \omega_{eq}$ (Fig. 4a), we assume that the gain of the amplifier is large enough that we can ignore the voltage swing at the FET gate. And we show later, distortion when $\omega_{blk} > \omega_{eq}$ is no longer important. In Fig. 14, the low-frequency small-signal voltage gain of the balanced CMOS inverter amplifier (assuming identical threshold voltage V_{t0} for both NFET and PFET) drops to 0 when the amplified blocker voltage $|v_{d,nclp}|$ exceeds the threshold voltage, $|v_{d,nclp}| > V_{t0}$. This happens over the period when $\omega_{blk}t \notin [-\phi_d, \phi_d] + n\pi,\ n \in \mathbb{Z}$, where $\phi_d = \arcsin(V_{t0}/V_{d0})$. It lowers the effective gain by a factor of $k_{avo} = 2\phi_d/\pi$. Specifically when V_d=B1dB, the gain has dropped by 1 dB $\simeq$ 10 %; therefore:

$$V_{d,B1dB} \approx \frac{V_{t0}}{\sin(0.446\pi)} \approx 1.01V_{t0} \simeq V_{t0}. \tag{47}$$

This result is consistent with the analysis in [25, Chap. 4].

(47) tells us that by using FETs with high threshold voltage, B1dB improves; [25, Chap. 4] recommends the same. This B1dB is smaller than predicted by the standard method for weak 3rd-order nonlinearity. Fig. 12b verifies the accuracy of this analysis.

At frequencies where the amplifier's voltage gain has rolled off, we may assume the opposite: a large voltage swing at the amplifier input (FET gates) results in a small voltage swing at the output (their drains). At the FET gate, that is, at the amplifier's *input port*, using the small-signal analysis (37)(40) and assuming weak nonlinearity, the B1dB lies ∼13 dB below IP3:

$$V_{g,B1dB,g30} \approx \sqrt{0.072\frac{g_m}{g_{30}}} \approx \sqrt{0.072\frac{2V_{P0}}{\theta}}. \tag{48}$$

In our circuit (Table II) $V_{g,B1dB,g30} \approx 0.38$ V, that is, 1.6 dBm. Since numerically $V_{g,B1dB,g30} \approx V_{P0}$; and since intuitively compression should appear when the peak of a sinewave momentarily turns off the FET, we conclude that (48) captures gain compression at the input port well enough that there is no further need for large-signal analysis. The nonlinearity in this case is weak.

Simulation of overall B1dB including contributions from nonlinearity at the output (drain) and input (gate) tracks analysis closely, as shown in Fig. 12b. The gate and drain will contribute equally to B1dB at the frequency

$$\omega_{B1dB} = \frac{V_{g,B1dB,g30}}{V_{d,B1dB}} \times \omega_{eq}. \tag{49}$$

When $\omega < \omega_{B1dB}$, gain compression is limited by output clipping. For $\omega > \omega_{B1dB}$, nonlinearity in g_m dominates. In our case, numerically $\omega_{B1dB} \approx \omega_{eq}$, consistent with simulation that ω_{eq} in Fig. 6 equals ω_{B1dB} in Fig. 12b. Whenever gain is compressed because the blocker input pushes the FETs from operating in saturation into triode, then ω_{B1dB} is $> \omega_{IP3,2,i}$ for $i \in [\![1, 2]\!]$.

V. Measurements

A prototype mixer-first receiver (Fig. 15a) is fabricated in Towerjazz 65-nm RF-SOI CMOS technology, as part of a monolithic electrical balance duplexer (EBD) [19] shown in Fig. 15b. Designed as an integral part of the duplexer, the RX operates between 5 GHz to 7 GHz in the Wi-Fi 6/6E bands. The passband of the TIA (BBBW) is 40 MHz wide to receive 80 MHz-wide RF channels at zero IF.

We now report measurements on the standalone receiver (Fig. 16) to illustrate the 2nd-order TIA design that was developed with the methodology described in this paper. An on-chip four-phase passive mixer drives the TIA. The mixer functions with RF inputs from 1 GHz to 7 GHz. The front-end takes 0.3 mm^2 of chip area.

A. Gain, s_{11}, Noise Figure

The TIA circuit (Fig. 16) uses capacitive positive feedback to obtain a maximally flat passband. The quasi-differential inverter OTA and its feedback elements follow the design guidelines of (26a) to (26e). C_F and C_2 realized as in Fig. 7a enable ω_0 and Q to be tuned independently. OTA g_m is 320 mS total, 80 mS in each path: this fulfills the design requirement that $g_m R_S \gg 1$. Ignoring FET capacitance, the switches in

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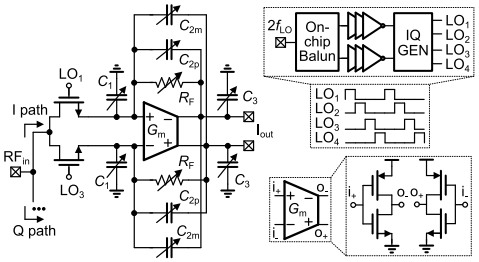


Fig. 16. Circuit diagram of the fabricated mixer-first receiver. (G_m cell NFET: 240/0.5µm; PFET: 700/0.5µm; R_F: 2.5-to-7.5 kΩ; C_1 & C_3: 0-to-6 pF; C_{2p} & C_{2m}: 2-to-9 pF (C_F tuning) plus 0-to-2.1 pF (C_2 tuning).)

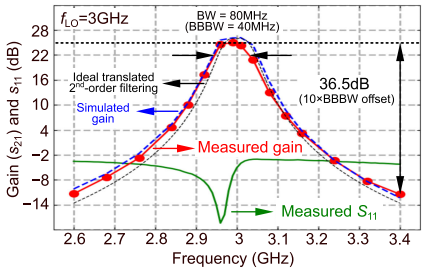


Fig. 17. Measured gain (s_{21}) and input reflection coefficient (s_{11}) of the fabricated mixer-first receiver.

each path tune $C_1 = C_2$ to 6 pF, C_{2p} to 8.7 pF, C_{2m} to 9.3 pF. According to (26), $f_0 = 38$ MHz, $Q = 0.89$, $f_{zg} = 28$ MHz, $f_{eq} = 530$ MHz ($\simeq 13\times$BBBW), $f_{zo} = 21$ GHz. No zero will appear at $|V_o/V_s(jf)|$ due to the high f_{zo}. In simulation, $Q \approx 1/\sqrt{2} = 0.707$ after accounting for FET C_{gd}. The inequality of (30) is satisfied during use, ensuring stable operation with large variations in antenna impedance. These design steps illustrate well the methodology of Sec. III-B.

Fig. 17 shows the measured frequency response with a 3 GHz LO driving the mixer. Measurement matches simulation. It also coincides with the simulated bandpass response formed by translating an ideal lowpass two-pole frequency response from DC[7]: this verifies that the complex mixer and TIA operate jointly as intended. Most importantly, it confirms that, by design, any zero in the measured transfer function $|V_o/V_s(jf)|$ lies beyond 400 MHz offset from the LO frequency (10×BBBW). The center frequency of this transfer function and of s_{11} is shifted slightly from the LO frequency of 3 GHz because of parasitic capacitance at the RF input [26], [27], [28].

Fig. 18a shows the measured gain and noise figure (NF) of the mixer-filter cascade as LO frequency is swept from 1 GHz to 7 GHz. Flicker noise is negligible: the simulated TIA flicker noise corner is about 500 kHz.

This SOI process was developed for high voltage use, so it is unable to realize competitive high-speed digital circuits. f_T of its low-voltage NFETs is ≈60 GHz, nearly 3× lower than of high-speed 65-nm bulk CMOS technologies. As clock frequency rises, the LO waveforms at the output of the on-chip four-phase clock generator that drives the mixers (Fig. 16) change gradually from full-swing square pulses into low-amplitude Gaussian-like pulses. As a result, both NF [26] and linearity [29] of the mixer suffer at high LO frequencies. A faster IC technology will certainly improve both [26], [29]. This degeneration at higher clock frequencies is seen in simulation, which reproduces the measurements. An analysis

[7]Whereas the skirts of a 2nd-order bandpass *LCR* resonator will roll off at 1st-order with frequency.

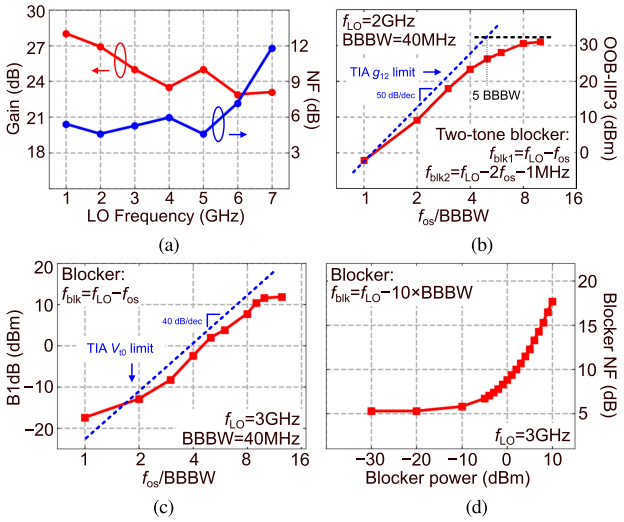


Fig. 18. Measured (a) gain/NF versus LO frequency, (b) OOB-IIP3 and (c) B1dB versus offset frequency, (d) block NF versus blocker power.

of how linearity and noise deteriorate in response to the shape of the clock waveform is beyond the scope of this paper.

B. Linearity and Blocker NF

The OOB-IIP3 is shown in Fig. 18b, measured at $f_{LO} = 2$ GHz. The input is two tones of equal amplitude at ($f_{LO} - f_{os}$) and ($f_{LO} - 2f_{os} - 1$ MHz). After downconversion to zero IF and encountering 3rd-order nonlinearity in the baseband circuits, or first suffering 3rd-order distortion at RF in the mixer and then downconversion to zero IF, these frequencies will produce an intermodulation-generated tone that always lies at 1 MHz for any f_{os}. Since this is in the filter's passband, it is measured straightforwardly.

For close-in blockers, theoretically the IIP3 of a 2nd-order TIA improves at 50 dB/dec with increasing f_{os}. The dominant source of distortion is $V_{IIP3,2,g_{12}}$, as discussed in Sec. IV-A.2. Eventually distortion in the mixer will pose an upper limit on IP3. When TIA filtering suppresses far-away blockers substantially, the mixer's distortion will limit IIP3 to some value that is nearly independent of f_{os}. In measurement (Fig. 18b), the onset of mixer distortion reveals itself at 5×BBBW. At offsets of 8×BBBW and beyond it dominates, limiting overall OOB-IIP3 to 31.1 dBm.

The results of analysis in (26) and (46) predict that when $g_m = 320$ mS, then $f_{eq} \simeq 13\times$BBBW, $f_{IP3,2,1} \simeq 1.8\times$BBBW, $f_{IP3,2,2} \simeq 5.6\times$BBBW and $f_{IP3,2,3} \simeq 6.8\times$BBBW. Referring to Fig. 12a, at low offset frequencies a *small* $f_{IP3,2,1}$ will limit the effects of distortion coefficient g_{03}. Whereas at high offset frequencies, *large* $f_{IP3,2,2}$ and $f_{IP3,2,3} > 5\times$BBBW imply that IIP3 due to the mixer will dominate the distortion arising from g_{21} and g_{30} in the TIA. This is the optimum design for blocker tolerance, because at every blocker frequency the bottleneck circuit is at its maximum possible IIP3.

The measurement in Fig. 18b shows that over different blocker offset frequencies, OOB-IIP3 is limited by (i) the distortion coefficient g_{12} of the amplifier at low frequencies, or (ii) the mixer, after the TIA has filtered the blocker substantially. We may claim therefore that this prototype circuit is well-optimized for linearity.

OOB-B1dB is shown in Fig. 18c when measured at $f_{LO} = 3$ GHz. OOB-IIP3, B1dB for close-in blockers improves at

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TABLE IV
MIXER-FIRST RECEIVER WITH HIGHER-ORDER BB FILTER COMPARISON TABLE

	JSSC'18 [7]	SSCL'21 [8]	MWSCAS'20 [9]	RFIC'18 [10]	JSSC'20 [11]	SSCL'21 [13]	JSSC'20 [14]	JSSC'20 [15]	This work
BB Architecture	cap. pos. feedback	reactance cancelling	neg. feedback *RC* network	shunting notch & cap. pos. feedback	2nd-order Z_{in}	4th-order Z_{in}	2nd-order NC TIA	CG-based 3rd-order TIA	cap. pos. feedback
Technology	45nm SOI	65nm	16nm (sim. only)	130nm SiGe BiCMOS	28nm	28nm	180nm	28nm	65nm SOI
Frequency (GHz)	0.2-8	0.5-2	0.2-3	2-11	0.2-2	0.2-3.5	0.2-1.2	0.5-2	1-7
Area (mm^2)	0.8	0.66	-	8.4	0.48	1.5	0.54	0.16	0.3
BBBW (MHz)	10	20	10	40-130	9	15	18	130	40
Gain (dB)	21	36	22.7	10-24	13	16.3	31.4	32.4	25
NF (dB)	2.3-5.4 (0.5-6GHz)	2.2-4.2	3.1 (1.5GHz)	11±1, 12±1† (5GHz)	4.3-7.6	6.6-11.7	3.4-4	5.5 (2GHz)	4.6-7.2 (1-6GHz)
IB-IIP3 (dBm)	-	-11	-	-2*, -2*†	4*	-	14.5	-14	-10
OOB-IIP3 ($\frac{f_{os}}{BBBW}$)	39dBm (8)	33dBm (11.4)	27dBm (9)	20dBm (1.75), 19dBm (2.5)†	33.3dBm (8.8)	22.5dBm (2)	39.8dBm (16*)	22dBm (4)	31.1dBm (10)
B1dB ($\frac{f_{os}}{BBBW}$)	12dBm (8)	14.5dBm (11.4)	13dBm (8)	1.8dBm (1.75), 1.2dBm (2.5)†	12dBm (6.6)	10dBm (4)	12dBm* (16*)	3dBm (6.2)	11.8dBm (12)
0 dBm Blocker NF Desense. (dB)	2.2	0*	0.9*	-	2	1.9	-	4.2	3.5
LO/BB Supply (V)	1.2	1.5/1.2	1.0	4.5/2.5, 4.0/2.5†	1.2	1.2/1.4	1.8	1.2/1.8	1.2
LO Power (mW)	30/GHz	32/GHz	13.9/GHz	1466-1494	3.6-36	2.5-62.5	45-135	7.8/GHz	211
BB Power (mW)	50	25	17.2	656, 674†	143	86	19.8	21.6	22

*: Estimated from figures.
†: shunting notch (former), and capacitive positive feedback (latter).

40 dB/dec with increasing f_{os}. And beyond ~10×BBBW, the mixer nonlinearity causes B1dB to saturate at 11.8 dBm. Our analysis (49) shows that the crossover frequency $f_{B1dB} \approx 13\times$BBBW. This means that the g_m nonlinearity g_{30} at the TIA's input is too weak to limit B1dB at any blocker frequency. The measurements of Fig. 18c support this: the absence of a 20 dB/dec region indicates that g_{30} is insignificant. It also indicates that the optimum design for linearity in this circuit is driven by considerations of IM3, not by gain compression.

Noise figure rises in the presence of a large blocker. Reciprocal mixing with LO phase noise, that is, when blocker spectral density at the mixer input convolves with the Lorentzian-shaped phase noise spectrum, creates a wide noise spectrum that is scaled-up by the blocker's power. This will encroach into the frequency band occupied by the channel being received, raising the RX NF. Fig. 18d shows the dependence of receiver NF on blocker power, where for convenience of measurement, the blocker is a swept-amplitude sinewave located at f_{os} of 10×BBBW. A 0 dBm sinewave blocker at this offset raises the measured NF by 3.5 dB. Gain compression was not responsible for this rise because B1dB is above 10 dBm.

C. Discussion

Table IV compares our chip results with other similar works on BB filters in mixer-first receivers. Our optimized TIA consumes less (BB) power compared to other circuits [7], [8], [10], including multi-OpAmp circuits [11], [13]. Our 4-phase LO generator consumes a large power because of the limitations of the SOI technology as described above.

When compared to [14], [15], our TIA circuit was developed using a simpler, cleaner design methodology and a well-decoupled method to tune ω_0 and Q of the 2nd-order frequency response. At comparable power, it gives optimum linearity. Fig. 12a and Fig. 18b illustrate what we mean by optimum design for blocker tolerance. As an example of non-optimum linearity, the receiver in [9] suffers from a low-frequency zero in the transfer functions arising from a feedback network with internal nodes (Fig. 3) that will limit both its OOB-IIP3 and B1dB to smaller values, as well as denying it the additional 10 dB/dec rate of rise in IIP3, a benefit that arises from the g_{12}-dominated region and filtering.

OOB-IIP3 of our circuit listed in Table IV is limited by the mixer nonlinearity, when the blocker modulates the FET switch on-resistance appreciably. The distortion is compounded by the deteriorated clock waveforms driving the mixer switches [26]. We expect that when switch FETs are turned on with a high gate voltage [8], [14] and clocks generated with fast FETs in more advanced technologies [7], [11], the circuit will display a higher OOB-IIP3. A differential topology [7] can lower on-resistance modulation of the mixer FETs to improve the mixer's IIP3, but at the cost of a balun.

Despite the handicap posed in this prototype by the clock drive to the mixer, we have satisfactorily verified the methodology of the baseband TIA. The circuit consumes low power and offers 2nd-order filtering with an optimized blocker dynamic range.

VI. CONCLUSION

This paper presents a detailed design analysis of a 2nd-order TIA. One preferred single-OpAmp TIA circuit topology is identified that offers second-order filtering over a wide frequency range. Simple and accurate design equations allow component tuning and selection in a meaningful sequence to meet specifications.

Analysis elaborates distortion mechanisms and shows how to achieve optimum blocker tolerance for both OOB-IIP3 and B1dB. Second-order filtering then improves linearity with blocker offset frequency until distortion in the front-end mixer prevails. We show the optimum design for blocker-tolerance. Power consumption (g_m) can be reduced until limited by noise and independent tuning requirements. A mixer-first receiver prototype is implemented. Consistent results among analysis, simulation, verify the correctness of the TIA analysis and design methodology.

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